CAS SciFinder

Task History

Initiating Search

November 4, 2025, 3:39 PM

Search:

CAS Lexicon Search:

Michael reaction - Preferred Concept

Michael reaction - Broader Concepts (1)

Addition reaction

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (51,576)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,464)	Reactions	View Results

Filtered By:

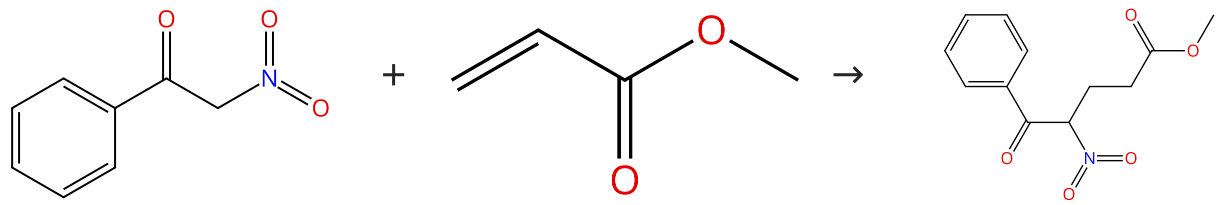
Yield:	**90-100%, 80-89%, 70-79%**
Document Type:	**Journal**
Language:	**English**
Publication Year:	**2020 to 2025**

CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (54)

Suppliers (60)

31-254-CAS-22778285 Steps: **1** Yield: **100%**

1.1 **Catalysts:** 18-Crown-6, Palladium(4+), tetrakis[μ-[3,3'-(1,3-phenylenedi-2,1-ethynediyl)bis[pyridine-κ*N*]]]di-, tetrakis[3,5-bis(trifluoromethyl)phenyl]borate(1-) (1:4)
Solvents: Dichloromethane-d_2; 88 h, rt

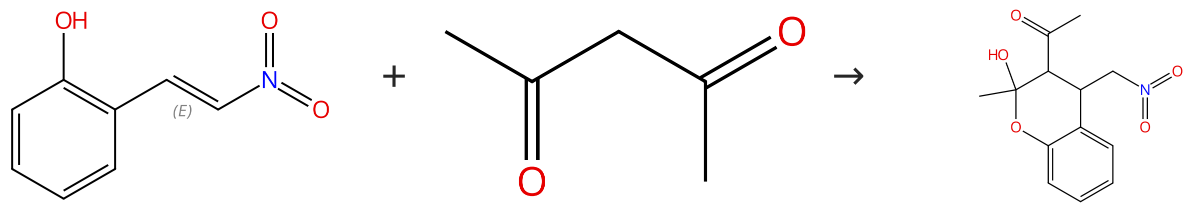
Experimental Protocols

Synergistic noncovalent catalysis facilitates base-free Michael addition

By: Wang, Jianzhu; et al

Journal of the American Chemical Society (2020), 142(41), 17743-17750.

Scheme 2 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (15)

Suppliers (76)

31-614-CAS-42835658 Steps: **1** Yield: **100%**

1.1 **Catalysts:** 3064750-60-0
Solvents: Isopropanol, Water; 6 h, rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

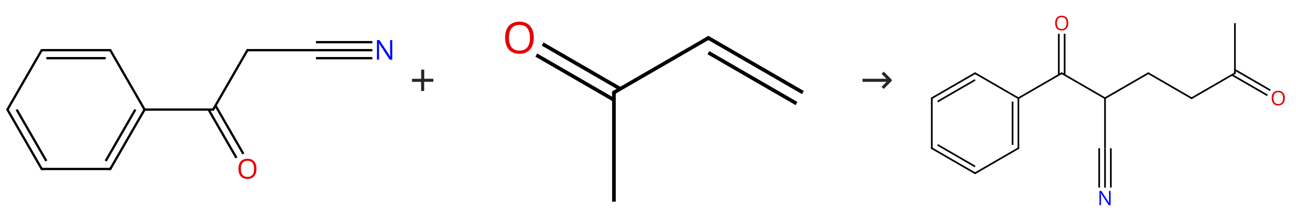
Experimental Protocols

Cascade Synthesis in Water: Michael Addition /Hemiketalization/Retro-Claisen Fragmentation Catalyzed by CatAnionic Vesicular Nanoreactor from Dithiocarbamate

By: Atibioke, Ayodele J.; et al

Chemistry - An Asian Journal (2025), 20(5), e202400853.

Scheme 3 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (92)

Suppliers (25)

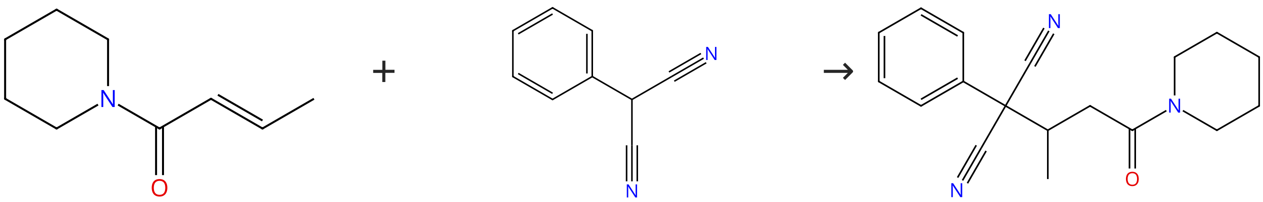
31-254-CAS-22778290	Steps: **1** Yield: **100%**
1.1	**Catalysts:** 18-Crown-6, Palladium(4+), tetrakis[μ-[3,3'-(1,3-phenylenedi-2,1-ethynediyl)bis[pyridine-κ*N*]]]di-, tetrakis[3,5-bis(trifluoromethyl)phenyl]borate(1-) (1:4) **Solvents:** Dichloromethane-d_2; 45 h, rt
Experimental Protocols	

Synergistic noncovalent catalysis facilitates base-free Michael addition

By: Wang, Jianzhu; et al

Journal of the American Chemical Society (2020), 142(41), 17743-17750.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (5)

Suppliers (52)

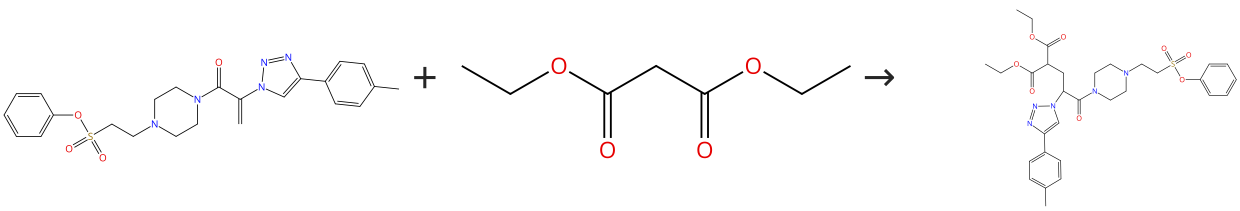
31-254-CAS-21750751	Steps: **1** Yield: **100%**
1.1	**Reagents:** 2-Methyl-2-butene **Catalysts:** Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, perchlorate (1:1) **Solvents:** Dichloromethane; 48 h, rt
Experimental Protocols	

Photoredox Michael addition of phenylmalononitrile onto α, β-unsaturated carboxylic acid

By: Hirahama, Toshiya; et al

Tetrahedron Letters (2020), 61(18), 151824.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (77)

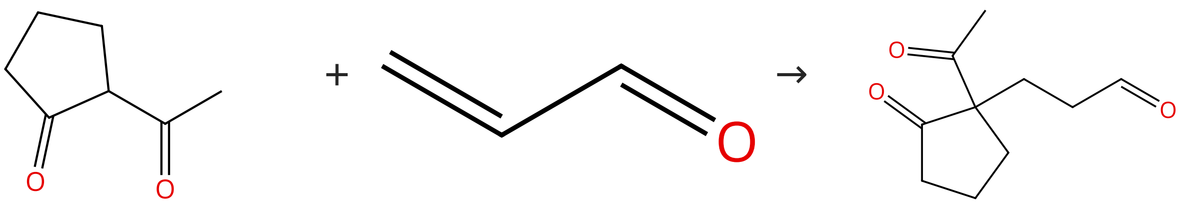
31-614-CAS-40396744	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium *tert*-butoxide **Solvents:** Tetrahydrofuran; 24 h, rt
Experimental Protocols	

Iterative click reactions using trivalent platforms for sequential molecular assembly

By: Orimoto, Gaku; et al

Chemical Communications (Cambridge, United Kingdom) (2024), 60(45), 5824-5827.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (60)

Suppliers (32)

31-254-CAS-22556281	Steps: **1** Yield: **100%**
1.1 **Solvents:** Tetrahydrofuran; 30 min, 100 °C	
Experimental Protocols	

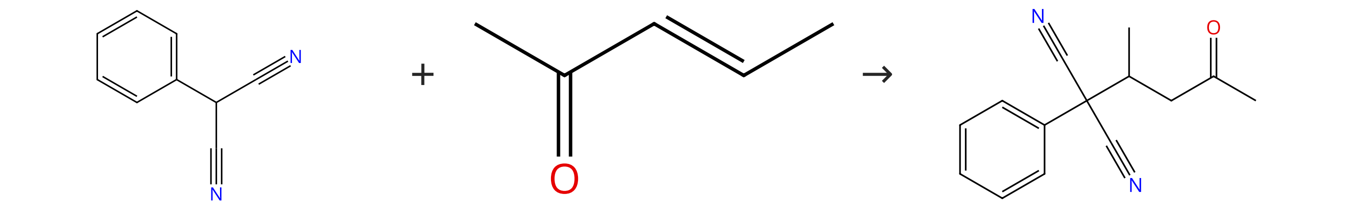
Michael Addition of 1,3-Dicarbonyl Derivatives in the Presence of Zeolite Y as an Heterogeneous Catalyst

By: Mekki, Amel; et al

Journal of Inorganic and Organometallic Polymers and Materials (2020), 30(7), 2323-2334.

Scheme 7 (1 Reaction)

Steps: **1** Yield: **100%**



31-254-CAS-21750749	Steps: **1** Yield: **100%**
1.1 **Reagents:** 2-Methyl-2-butene **Catalysts:** Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, perchlorate (1:1) **Solvents:** Dichloromethane; 48 h, rt	
Experimental Protocols	

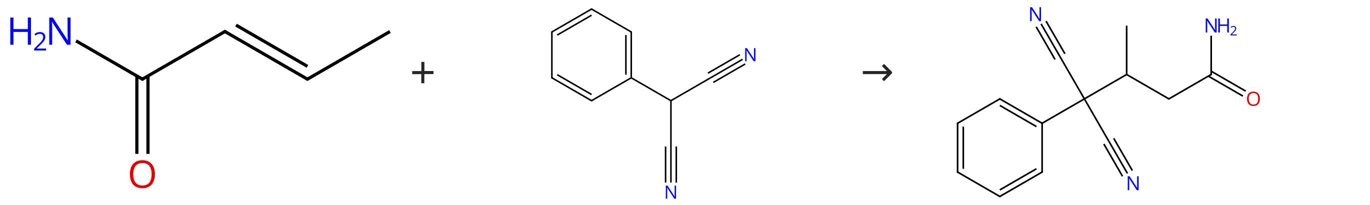
Photoredox Michael addition of phenylmalo nonitrile onto α, β-unsaturated carboxylic acid

By: Hirahama, Toshiya; et al

Tetrahedron Letters (2020), 61(18), 151824.

Scheme 8 (1 Reaction)

Steps: **1** Yield: **100%**



31-254-CAS-21750750	Steps: **1** Yield: **100%**
1.1 **Reagents:** 2-Methyl-2-butene **Catalysts:** Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, perchlorate (1:1) **Solvents:** Dimethylformamide; 48 h, rt	
Experimental Protocols	

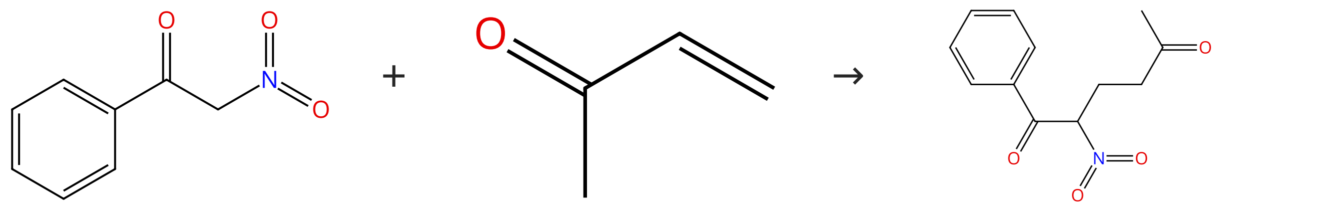
Photoredox Michael addition of phenylmalo nonitrile onto α, β-unsaturated carboxylic acid

By: Hirahama, Toshiya; et al

Tetrahedron Letters (2020), 61(18), 151824.

Scheme 9 (1 Reaction)

Steps: **1** Yield: **100%**



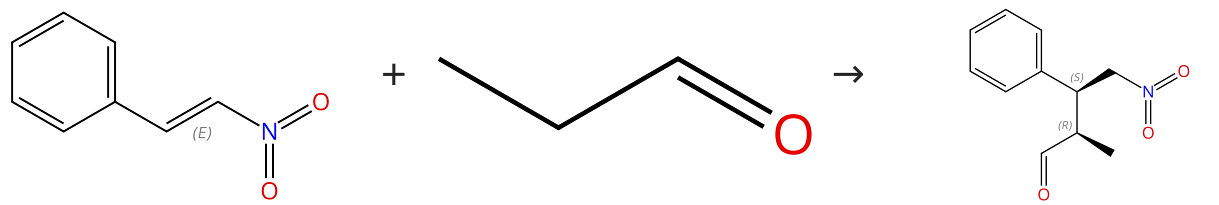
31-254-CAS-22778284	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Palladium(4+), tetrakis[μ-[3,3'-(1,3-phenylenedi-2,1-ethynediyl)bis[pyridine-κ*N*]]]di-, tetrakis[3,5-bis(trifluoromethyl)phenyl]borate(1-) (1:4) **Solvents:** Dichloromethane-d_2; 111 h, rt
Experimental Protocols	

Synergistic noncovalent catalysis facilitates base-free Michael addition

By: Wang, Jianzhu; et al

Journal of the American Chemical Society (2020), 142(41), 17743-17750.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (91)

Suppliers (75)

Relative stereochemistry shown

Suppliers (3)

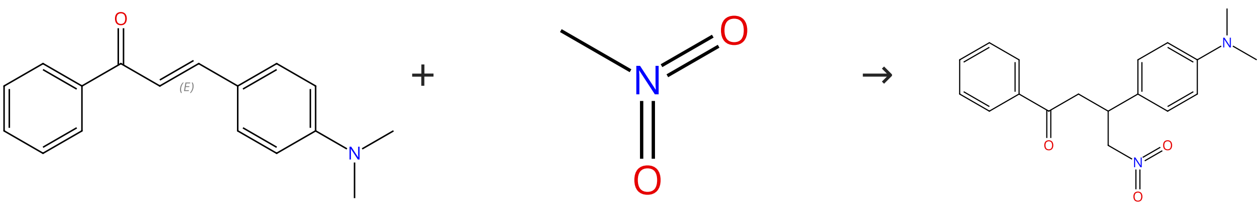
31-614-CAS-31409278	Steps: **1** Yield: **100%**
1.1	**Catalysts:** 2758369-85-4 **Solvents:** Toluene; 5 h, rt
Experimental Protocols	

Amphiphilic Immobilized Diphenylprolinol Alkyl Ether Catalyst on PS-PEG Resin

By: Koshino, Seitaro; et al

Bulletin of the Chemical Society of Japan (2021), 94(3), 790-797.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (44)

Suppliers (92)

Suppliers (3)

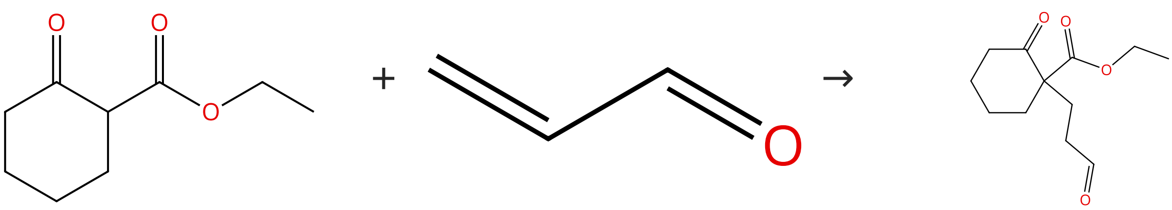
31-254-CAS-21680491	Steps: **1** Yield: **100%**
1.1	**Reagents:** Diethylamine **Catalysts:** 1,8-Diazabicyclo[5.4.0]undec-7-ene; 24 h, rt

1,2,4-Triphenylpyrroles: Synthesis, Structure and Luminescence Properties

By: Ferreira, Joana R. M.; et al

Synlett (2020), 31(6), 632-634.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (85)

Suppliers (32)

Suppliers (2)

31-254-CAS-22556283	Steps: **1** Yield: **100%**
1.1 **Solvents:** Tetrahydrofuran; 45 min, 100 °C	
Experimental Protocols	

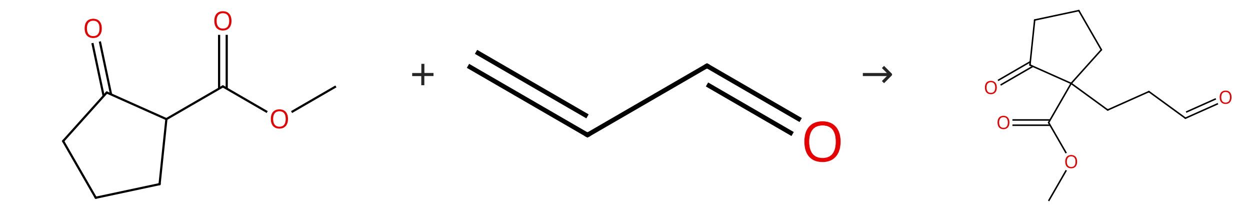
Michael Addition of 1,3-Dicarbonyl Derivatives in the Presence of Zeolite Y as an Heterogeneous Catalyst

By: Mekki, Amel; et al

Journal of Inorganic and Organometallic Polymers and Materials (2020), 30(7), 2323-2334.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (92) Suppliers (32) Suppliers (5)

31-254-CAS-22556279	Steps: **1** Yield: **100%**
1.1 **Solvents:** Tetrahydrofuran; 15 min, 100 °C	
Experimental Protocols	

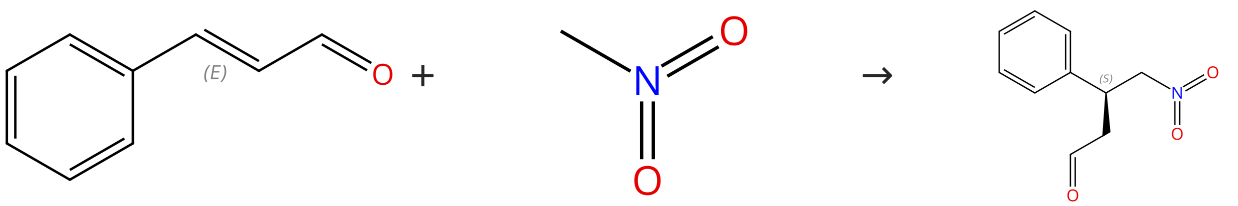
Michael Addition of 1,3-Dicarbonyl Derivatives in the Presence of Zeolite Y as an Heterogeneous Catalyst

By: Mekki, Amel; et al

Journal of Inorganic and Organometallic Polymers and Materials (2020), 30(7), 2323-2334.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **100%**



Double bond geometry shown

Absolute stereochemistry shown, Rotation (-)

Suppliers (109) Suppliers (92) Suppliers (3)

31-614-CAS-33503923	Steps: **1** Yield: **100%**
1.1 **Reagents:** Benzoic acid **Catalysts:** (3*R*,5*S*)-5-[Diphenyl[(trimethylsilyl)oxy]methyl]-3-pyrrolidinol (polymer-bound) **Solvents:** Methanol, Water; 3 h, 65 °C	
Experimental Protocols	

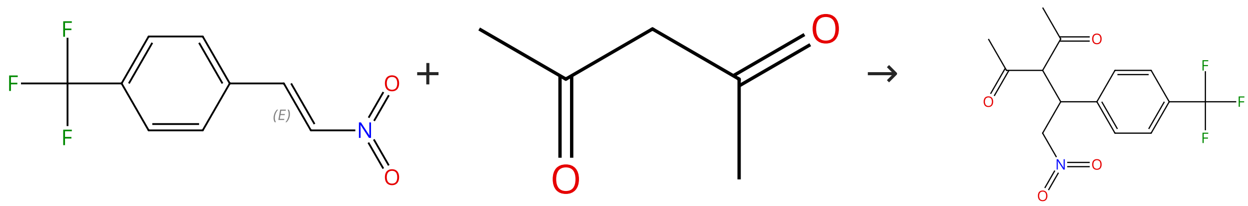
Asymmetric flow reactions catalyzed by immobi lized diphenyl prolinol alkyl ether: Michael reaction and domino reactions

By: Hayashi, Yujiro; et al

Chemistry - An Asian Journal (2022), 17(14), e202200314.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (52) Suppliers (76)

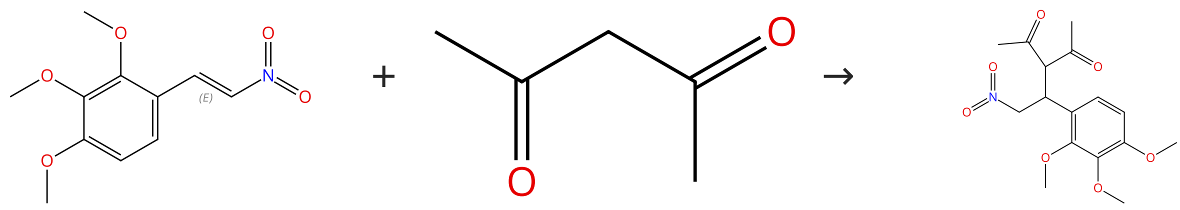
31-254-CAS-21265685	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 2377799-78-3 **Solvents:** Ethanol; 16 h, rt
Experimental Protocols	

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 16 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (19)

Suppliers (76)

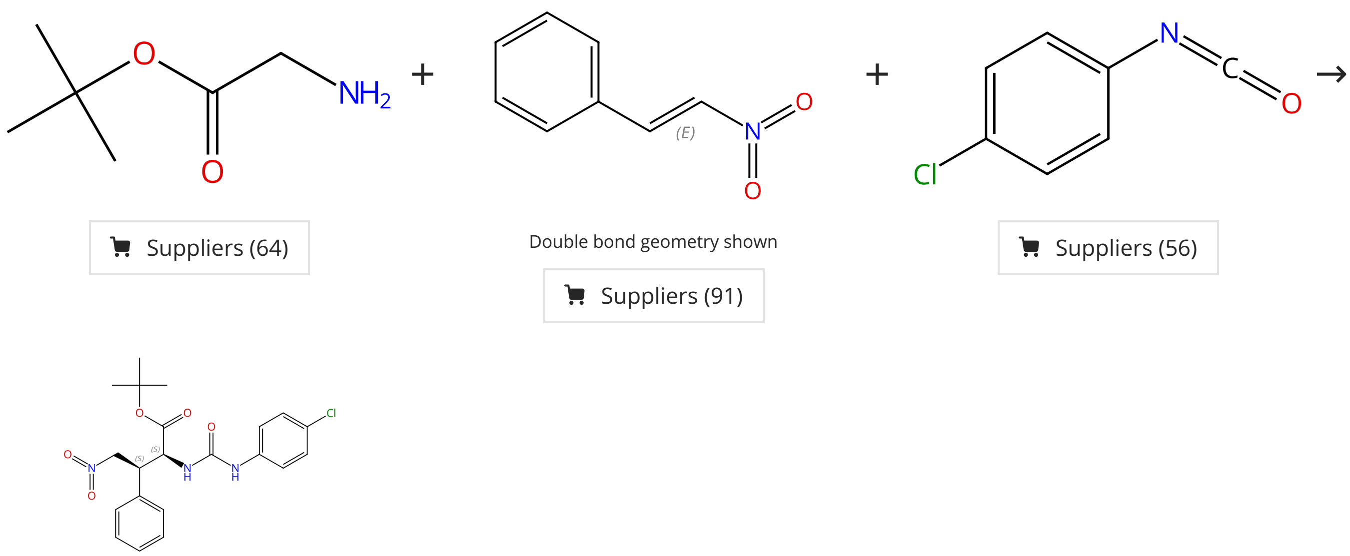
31-254-CAS-21265694	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 2377799-78-3 **Solvents:** Ethanol; 16 h, rt
Experimental Protocols	

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 17 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (64)

Double bond geometry shown

Suppliers (91)

Suppliers (56)

Absolute stereochemistry shown, Rotation (-)

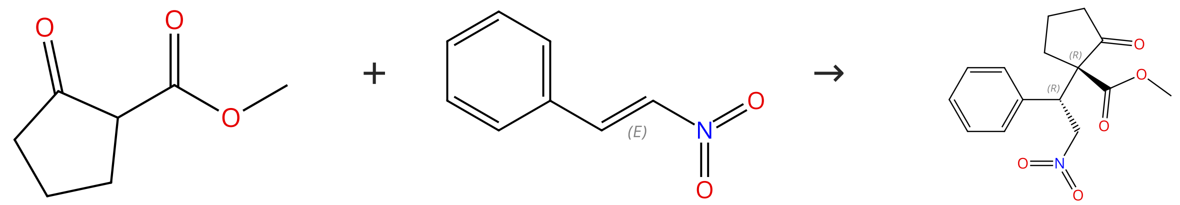
31-614-CAS-46206367	Steps: **1** Yield: **99%**
1.1	**Solvents:** Dichloromethane; 4 h, rt
Experimental Protocols	

Direct Asymmetric α-C Addition of Glycinate to Nitroalkenes by Pyridoxal/Nd Dual Catalysis

By: Si, Shibo; et al

Angewandte Chemie, International Edition (2025), 64(34), e202506342.

Scheme 18 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (92)

Double bond geometry shown

Suppliers (91)

Relative stereochemistry shown

31-614-CAS-24779958 Steps: **1** Yield: **99%**

1.1 **Catalysts:** *N*-[3,5-Bis(trifluoromethyl)phenyl]-*N*′-[(1*R*,2*R*)-2-[[[3,5-bis(trifluoromethyl)phenyl]methyl]amino]-1,2-diphenylethyl]thiourea
Solvents: Water; 24 h, rt

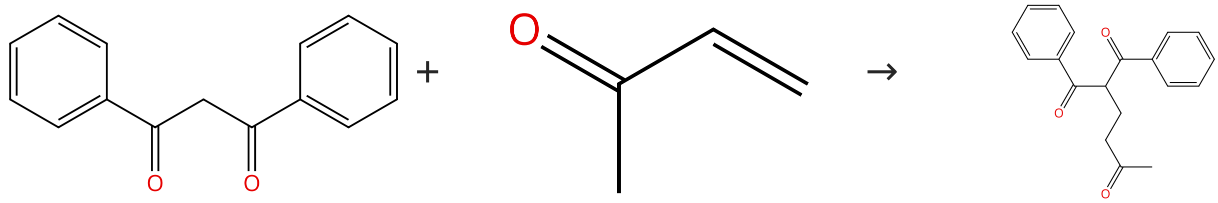
Experimental Protocols

Organocatalytic Asymmetric Michael Addition in Aqueous Media by a Hydrogen-Bonding Catalyst and Application for Inhibitors of $GABA_B$ Receptor

By: Shim, Jae Ho; et al

Catalysts (2021), 11(9), 1134.

Scheme 19 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (94)

Suppliers (25)

Supplier (1)

31-254-CAS-22315553 Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 48 h, 90 °C

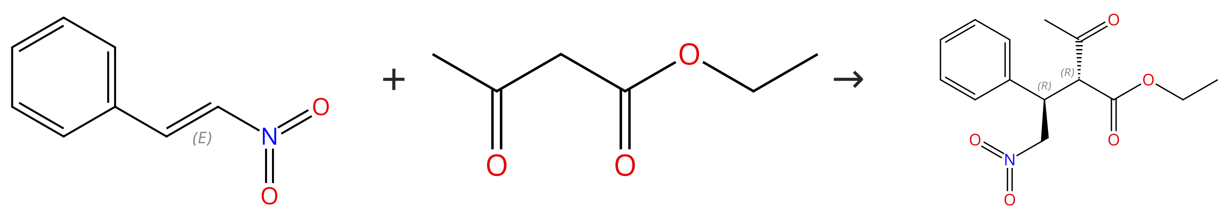
Experimental Protocols

Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 20 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (91)

Suppliers (96)

Relative stereochemistry shown

Supplier (1)

31-614-CAS-24779965 Steps: **1** Yield: **99%**

1.1 **Catalysts:** *N*-[3,5-Bis(trifluoromethyl)phenyl]-*N*′-[(1*R*,2*R*)-2-[[[3,5-bis(trifluoromethyl)phenyl]methyl]amino]-1,2-diphenylethyl]thiourea
Solvents: Water; 24 h, rt

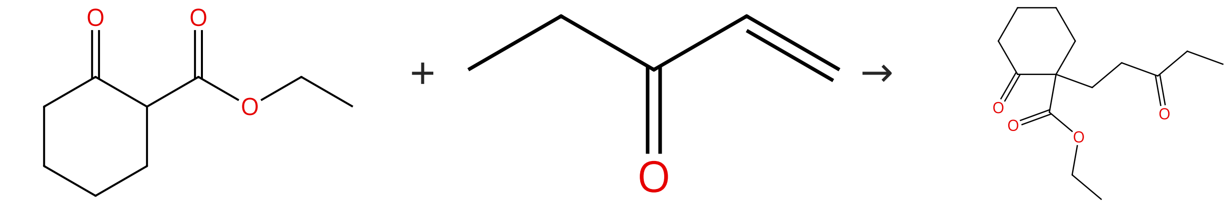
Experimental Protocols

Organocatalytic Asymmetric Michael Addition in Aqueous Media by a Hydrogen-Bonding Catalyst and Application for Inhibitors of $GABA_B$ Receptor

By: Shim, Jae Ho; et al

Catalysts (2021), 11(9), 1134.

Scheme 21 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (85) Suppliers (33) Supplier (1)

31-254-CAS-22315555 Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 64 h, 90 °C

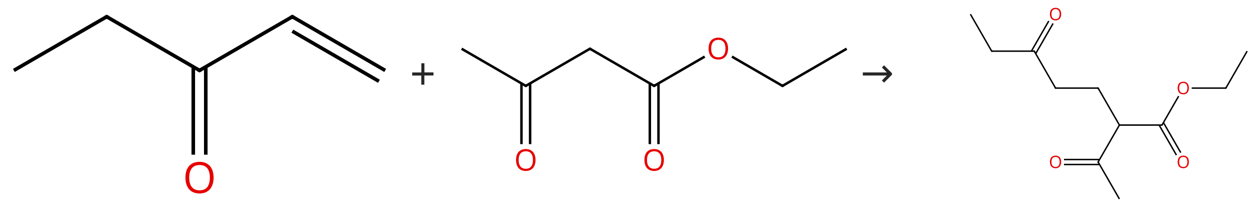
Experimental Protocols

Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 22 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (33) Suppliers (96) Suppliers (2)

31-254-CAS-22315556 Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 36 h, 90 °C

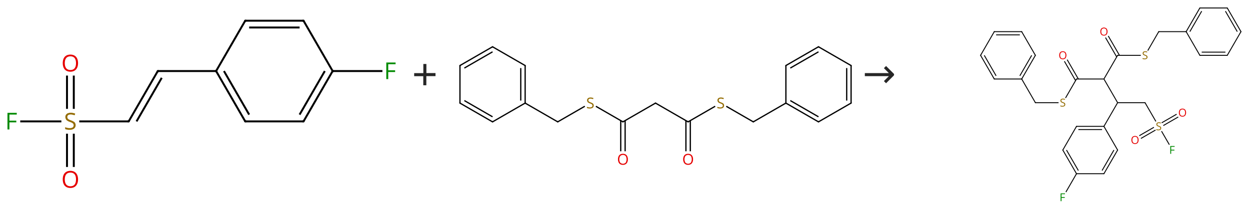
Experimental Protocols

Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 23 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (11) Suppliers (2)

31-614-CAS-35532708 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium chloride
Catalysts: *N*'''-[*P*,*P*-Bis(dimethylamino)-*N*-(1,1-dimethylethyl)phosphinimyl]-*N*,*N*,*N*',*N*',*N*'',*N*''-hexamethylphosphorimidic triamide
Solvents: Toluene, Water; 30 h, 0 °C

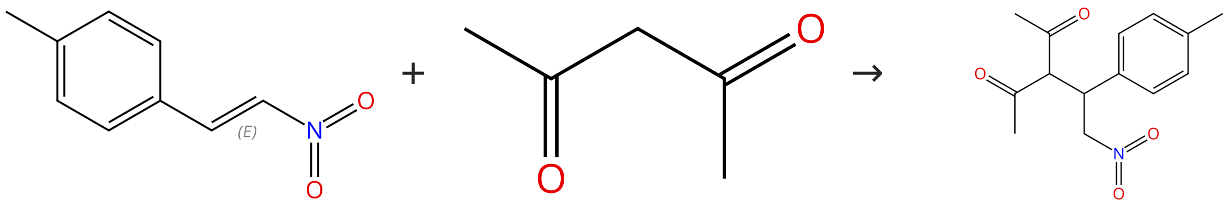
Experimental Protocols

Alkyl Sulfonyl Fluorides Incorporating Geminal Dithioesters as SuFEx Click Hubs via Water-Accelerated Organosuperbase Catalysis

By: Park, Jin Hyun; et al

Organic Letters (2023), 25(7), 1056-1060.

Scheme 24 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (70)

Suppliers (76)

Supplier (1)

31-254-CAS-21265683 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2377799-78-3
Solvents: Ethanol; 16 h, rt

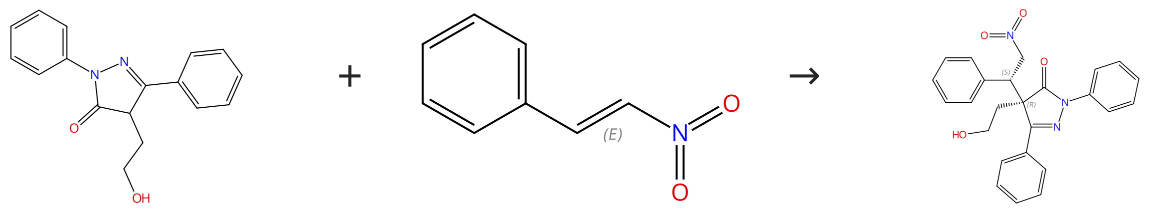
Experimental Protocols

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 25 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (2)

Double bond geometry shown

Suppliers (91)

31-614-CAS-37322761 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2973315-00-1
Solvents: Dichloromethane; 2 d, 25 °C

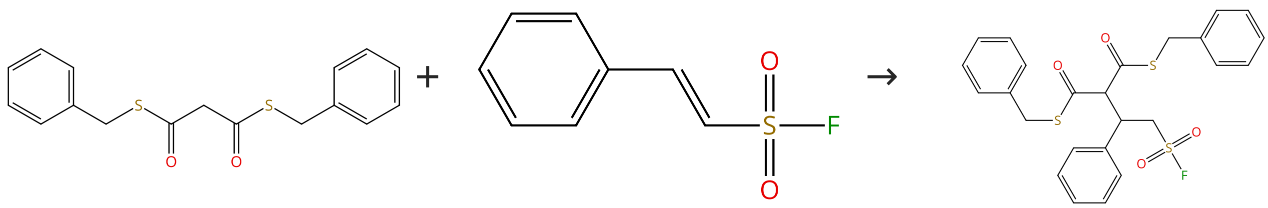
Experimental Protocols

Cooperative Lewis Acid-1,2,3-Triazolium-Aryloxide Catalysis: Pyrazolone Addition to Nitroolefins as Entry to Diamino amides

By: Wanner, Daniel M.; et al

Angewandte Chemie, International Edition (2023), 62(36), e202307317.

Scheme 26 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (2)

Suppliers (31)

31-614-CAS-35532716 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium chloride
Catalysts: *N*'''-[*P*,*P*-Bis(dimethylamino)-*N*-(1,1-dimethylethyl) phosphinimyl]-*N*,*N*,*N*',*N*',*N*'',*N*''-hexamethylphosphorimidic triamide
Solvents: Toluene, Water; 30 h, 0 °C

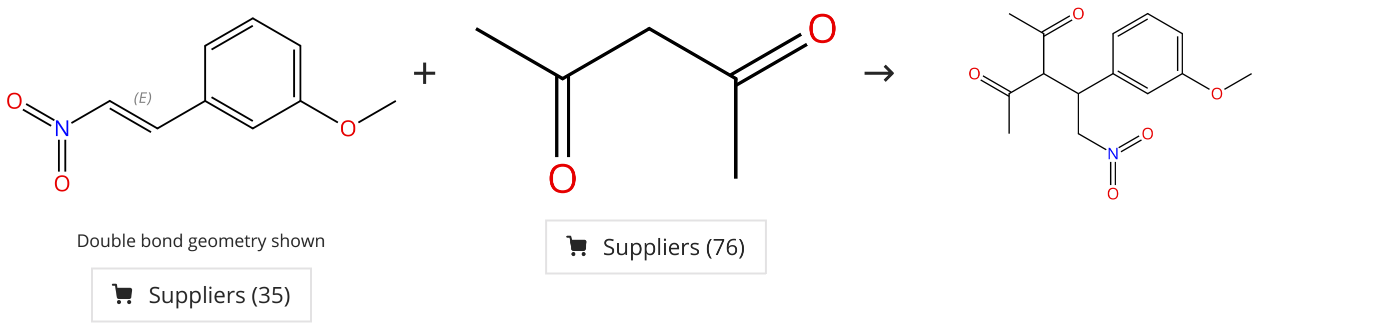
Experimental Protocols

Alkyl Sulfonyl Fluorides Incorporating Geminal Dithioesters as SuFEx Click Hubs via Water-Accelerated Organosuperbase Catalysis

By: Park, Jin Hyun; et al

Organic Letters (2023), 25(7), 1056-1060.

Scheme 27 (1 Reaction) Steps: **1** Yield: **99%**



31-254-CAS-21265691 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2377799-78-3
Solvents: Ethanol; 16 h, rt

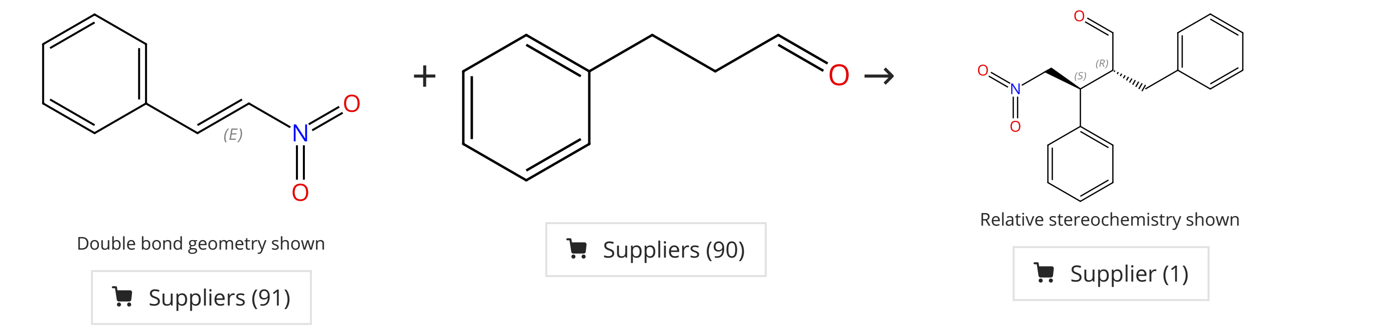
Experimental Protocols

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 28 (1 Reaction) Steps: **1** Yield: **99%**



31-614-CAS-31409277 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (2*S*)-2-[(2-Naphthalenylmethoxy)diphenylmethyl] pyrrolidine
Solvents: Toluene; 4 h, rt

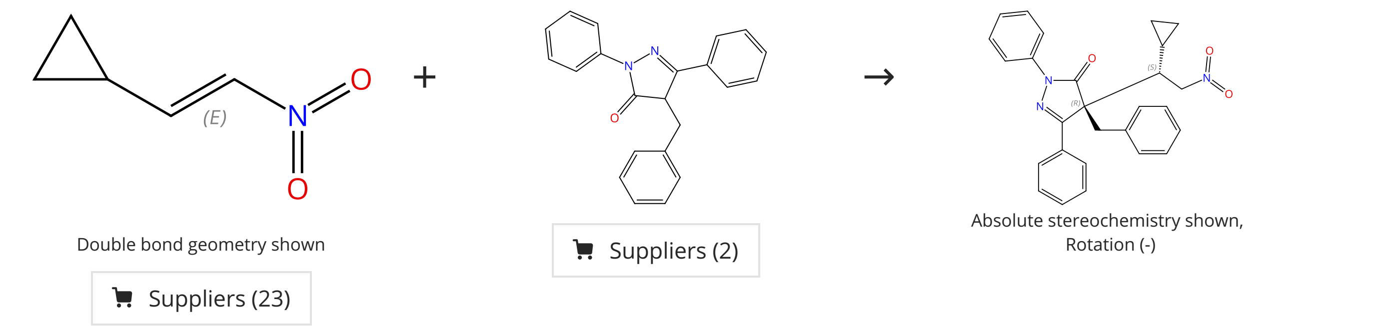
Experimental Protocols

Amphiphilic Immobilized Diphenylprolinol Alkyl Ether Catalyst on PS-PEG Resin

By: Koshino, Seitaro; et al

Bulletin of the Chemical Society of Japan (2021), 94(3), 790-797.

Scheme 29 (1 Reaction) Steps: **1** Yield: **99%**



31-614-CAS-37322769 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2973315-00-1
Solvents: Dichloromethane; 2 d, 25 °C

Experimental Protocols

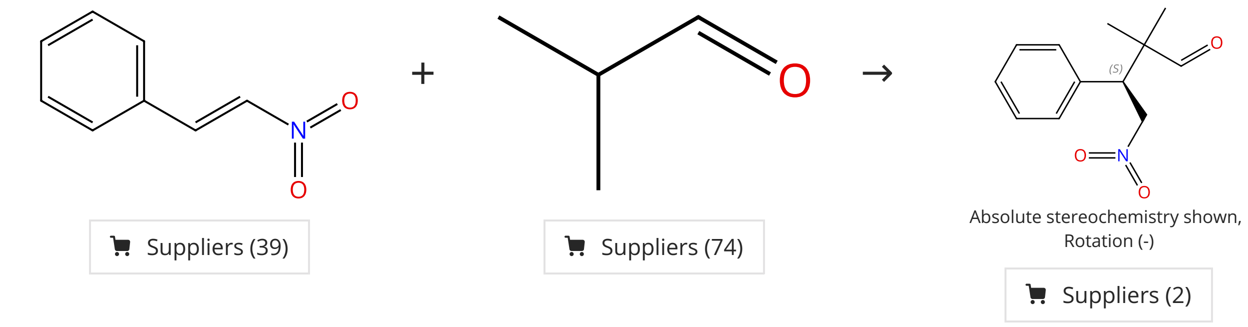
Cooperative Lewis Acid-1,2,3-Triazolium-Aryloxide Catalysis: Pyrazolone Addition to Nitroolefins as Entry to Diamino amides

By: Wanner, Daniel M.; et al

Angewandte Chemie, International Edition (2023), 62(36), e202307317.

Scheme 30 (1 Reaction)

Steps: **1** Yield: **99%**



31-614-CAS-31744832

Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2766488-98-4
Solvents: Dichloromethane; 10 min, 20 °C

1.2 4 d, 20 °C

Experimental Protocols

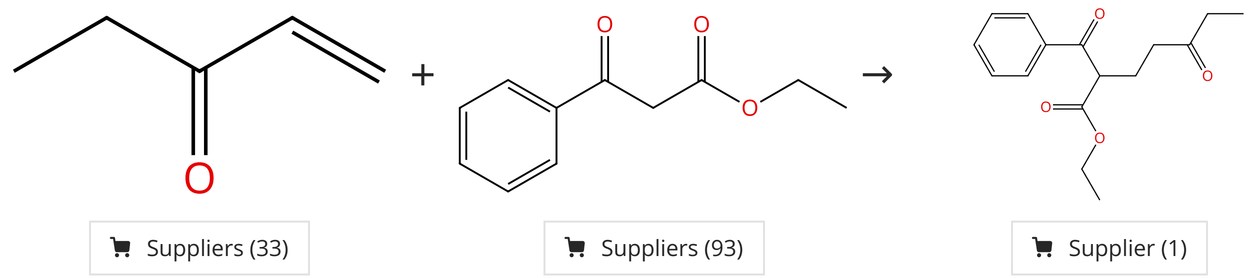
Chiral C_2-symmetric bis-thioureas as enzyme mimics in enantioselective Michael addition

By: Cruz, Harold; et al

Chirality (2022), 34(6), 877-886.

Scheme 31 (1 Reaction)

Steps: **1** Yield: **99%**



31-254-CAS-22315557

Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 64 h, 90 °C

Experimental Protocols

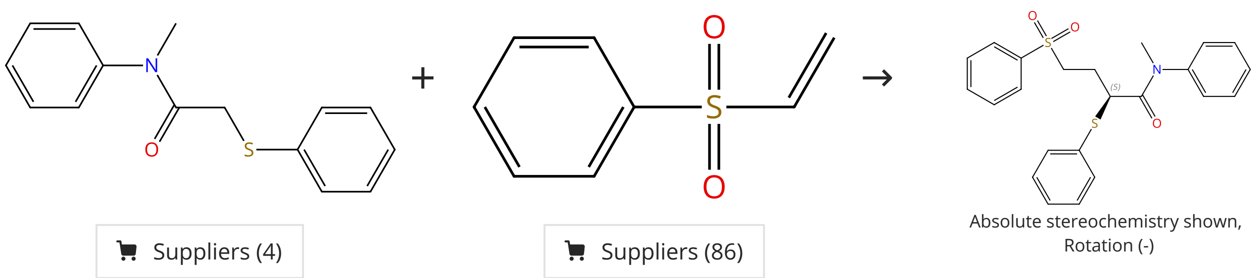
Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **99%**



31-254-CAS-21587983

Steps: **1** Yield: **99%**

1.1 **Catalysts:** Sodium *tert*-butoxide, 2411743-13-8
Solvents: Toluene; 10 min, rt

1.2 5 min, rt

1.3 30 min, rt

1.4 **Reagents:** Ammonium chloride
Solvents: Water; rt

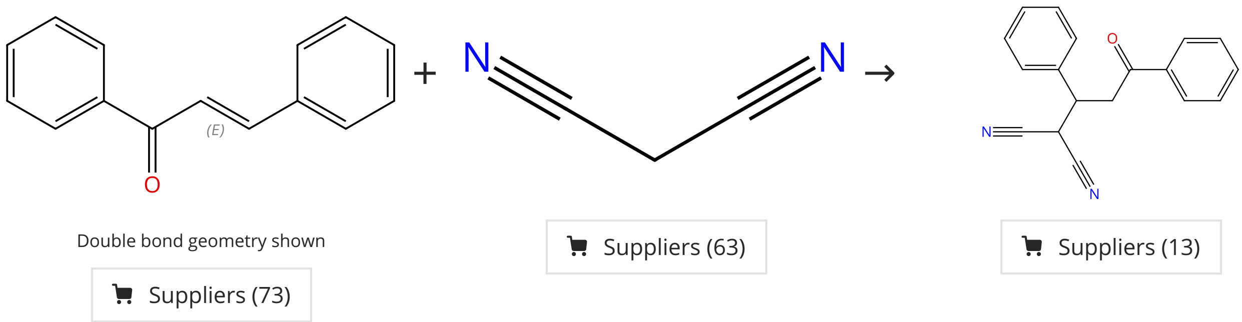
Development of Chiral Ureates as Chiral Strong Bronsted Base Catalysts

By: Kondoh, Azusa; et al

Journal of the American Chemical Society (2020), 142(8), 3724-3728.

Scheme 33 (1 Reaction)

Steps: **1** Yield: **99%**



31-614-CAS-37486615	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Urea, Choline chloride, L-Proline, Dibenzylidene sorbitol; 24 h, 25 °C

Experimental Protocols

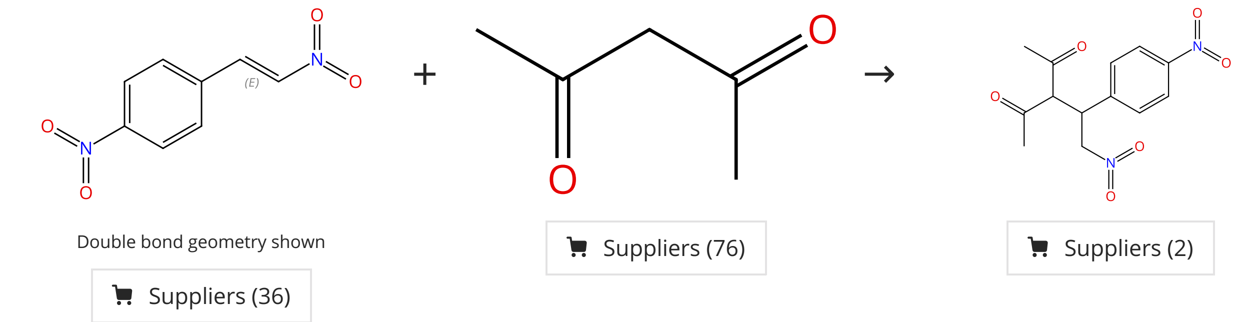
DBS-Based Eutectogels: Organized Vessels to Perform the Michael Addition Reaction

By: Rizzo, Carla; et al

European Journal of Organic Chemistry (2023), 26(35), e202300263.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **99%**



31-254-CAS-21265686	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 2377799-78-3 **Solvents:** Ethanol; 16 h, rt

Experimental Protocols

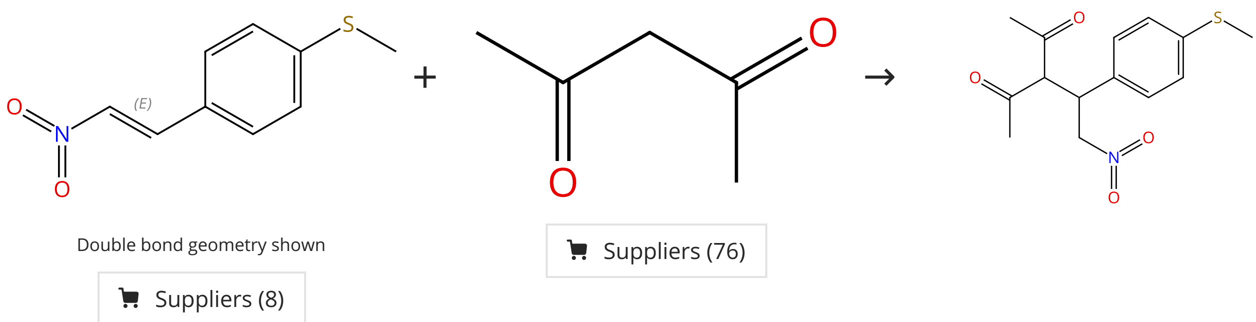
Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **99%**



31-254-CAS-21265684	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 2377799-78-3 **Solvents:** Ethanol; 16 h, rt

Experimental Protocols

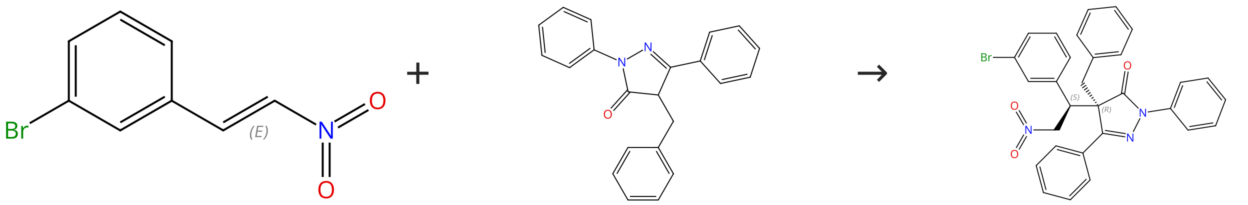
Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (47)

Suppliers (2)

Absolute stereochemistry shown, Rotation (-)

31-614-CAS-37322772

Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2973315-00-1
Solvents: Dichloromethane; 2 d, 25 °C

Experimental Protocols

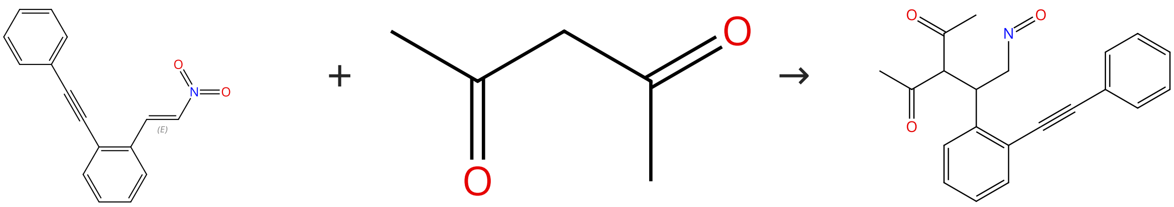
Cooperative Lewis Acid-1,2,3-Triazolium-Aryloxide Catalysis: Pyrazolone Addition to Nitroolefins as Entry to Diamino amides

By: Wanner, Daniel M.; et al

Angewandte Chemie, International Edition (2023), 62(36), e202307317.

Scheme 37 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (2)

Suppliers (76)

31-614-CAS-36892847

Steps: **1** Yield: **99%**

1.1 **Catalysts:** 3-Cyclobutene-1,2-dione, 3-[[(1*R*,2*R*)-2-(dimethylamino)cyclohexyl]amino]-4-[[(1*S*)-1,2,3,4-tetrahydro-1-naphthalenyl]amino]-
Solvents: Toluene; 24 h, 20 °C

Experimental Protocols

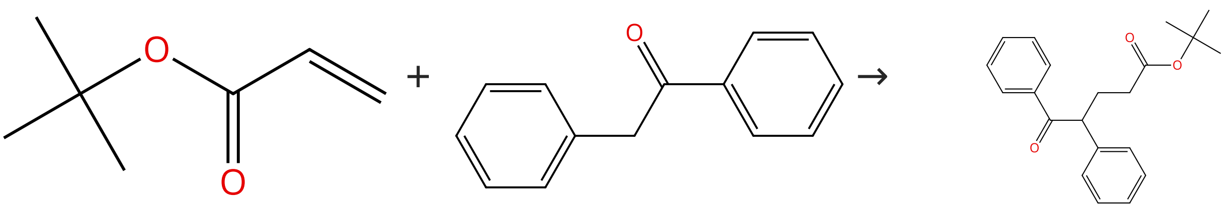
The Chameleonic Nature of the Nitro Group Applied to a Base-Promoted Cascade Reaction To Afford Indane-Fused Dihydrofurans

By: Diaz-Salazar, Howard; et al

Journal of Organic Chemistry (2023), 88(13), 8150-8162.

Scheme 38 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (56)

Suppliers (98)

31-614-CAS-40570413

Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2-[3,5-Bis(trifluoromethyl)phenyl]-2-methyloxirane, Cyclopropanecarboxylic acid, 1-phenyl-, compd. with 2-methyl-*N*-(tri-1-pyrrolidinylphosphoranylidene)-2-propanamine (1:1)
Solvents: Dimethyl sulfoxide; 24 h, 25 °C

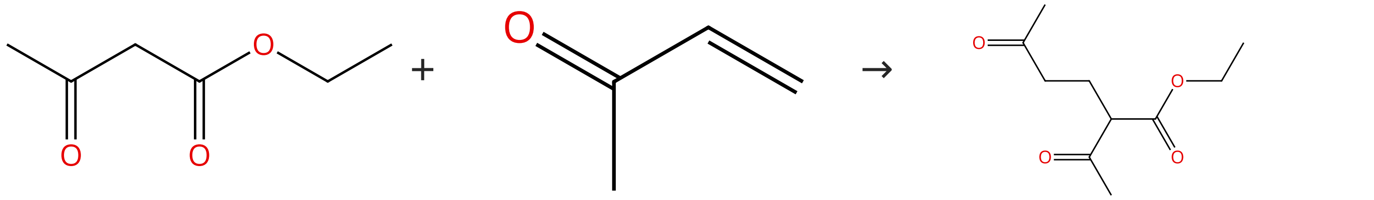
Experimental Protocols

A strategy for the contro llable generation of organic superbases from benchtop-stable salts

By: Sujansky, Stephen J.; et al

Chemical Science (2024), 15(26), 10018-10026.

Scheme 39 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (96) Suppliers (25) Suppliers (12)

31-254-CAS-22315547 Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 8 h, 90 °C

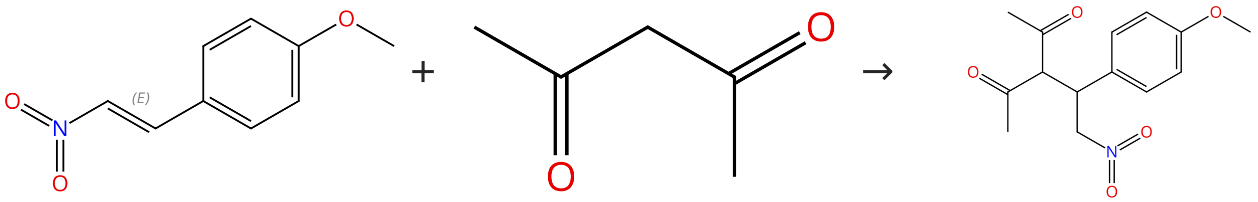
Experimental Protocols

Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 40 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (61) Suppliers (76) Suppliers (2)

31-254-CAS-21265690 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2377799-78-3
Solvents: Ethanol; 16 h, rt

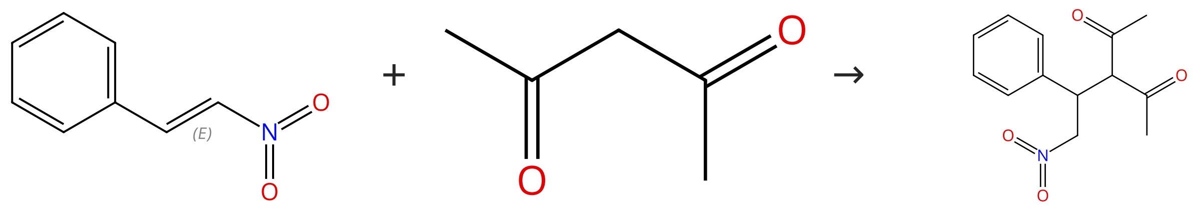
Experimental Protocols

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 41 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (91) Suppliers (76) Suppliers (6)

31-254-CAS-21265682 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2377799-78-3
Solvents: Ethanol; 16 h, rt

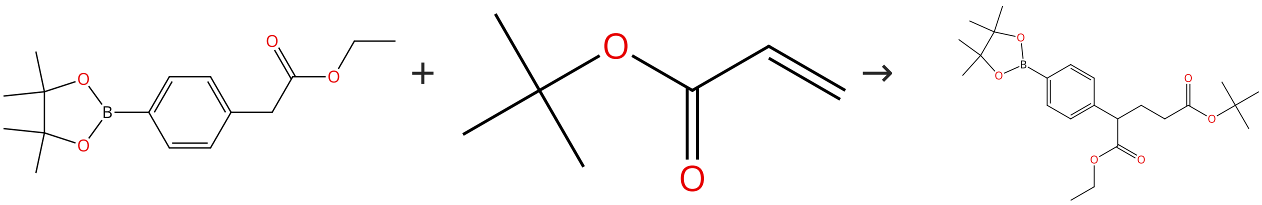
Experimental Protocols

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.

Scheme 42 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (80) Suppliers (56)

31-614-CAS-40570416 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2-[3,5-Bis(trifluoromethyl)phenyl]-2-methyloxirane, Cyclopropanecarboxylic acid, 1-phenyl-, compd. with 2-methyl-*N*-(tri-1-pyrrolidinylphosphoranylidene)-2-propanamine (1:1)
Solvents: Dimethyl sulfoxide; 24 h, 120 °C

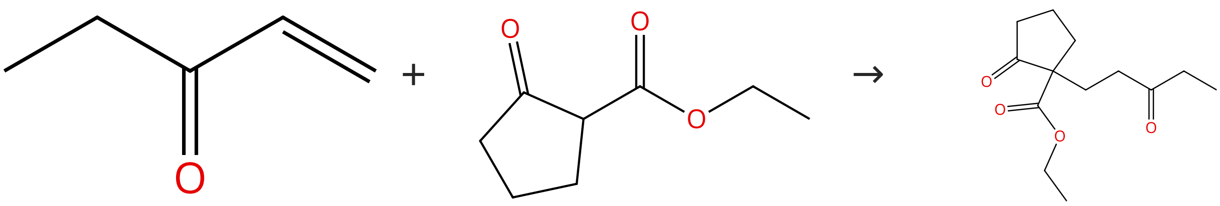
Experimental Protocols

A strategy for the contro llable generation of organic superbases from benchtop-stable salts

By: Sujansky, Stephen J.; et al

Chemical Science (2024), 15(26), 10018-10026.

Scheme 43 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (33) Suppliers (83) Suppliers (4)

31-254-CAS-22315554 Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 24 h, 70 °C

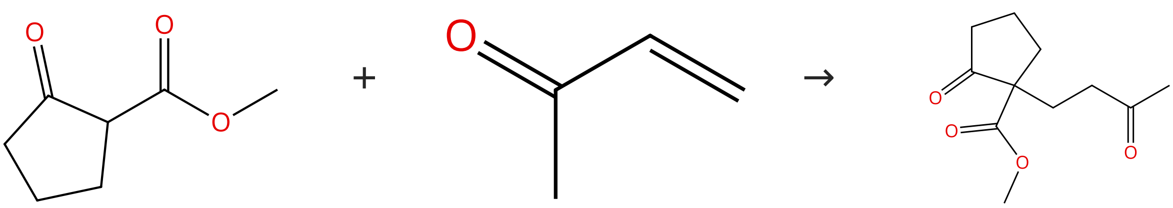
Experimental Protocols

Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 44 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (92) Suppliers (25) Suppliers (4)

31-254-CAS-22315543 Steps: **1** Yield: **99%**

1.1 **Reagents:** Silica; 0 °C; 8 h, 70 °C

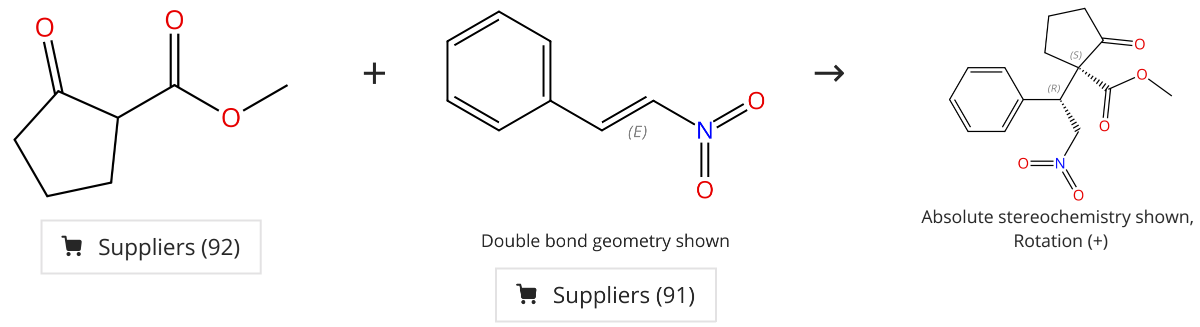
Experimental Protocols

Silica gel-mediated catalyst-free and solvent-free Michael addition of 1,3-dicarbonyl compounds to highly toxic methyl vinyl ketone without volatilization

By: Tanemura, Kiyoshi; et al

Tetrahedron Letters (2020), 61(30), 152142.

Scheme 45 (1 Reaction) Steps: **1** Yield: **99%**



31-614-CAS-37446697 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Sorbitol, Betaine, Water, 3-[[[3,5-Bis(trifluoromethyl)phenyl]methyl]amino]-4-[(8α,9*S*)-cinchonan-9-ylamino]-3-cyclobutene-1,2-dione; 24 h, 40 °C

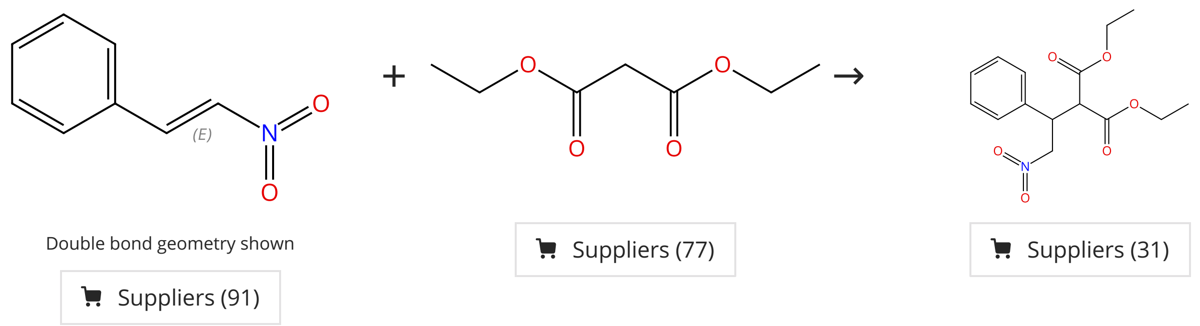
Experimental Protocols

Sustainable Organocatalyzed Enantioselective Catalytic Michael Additions in Betaine-Derived Deep Eutectic Solvents

By: Fonseca, Daniela P.; et al

SynOpen (2023), 7(3), 374-380.

Scheme 46 (1 Reaction) Steps: **1** Yield: **99%**



31-254-CAS-21933601 Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylamine
Catalysts: 2415283-23-5
Solvents: Dichloromethane-d_2; 12 h, rt

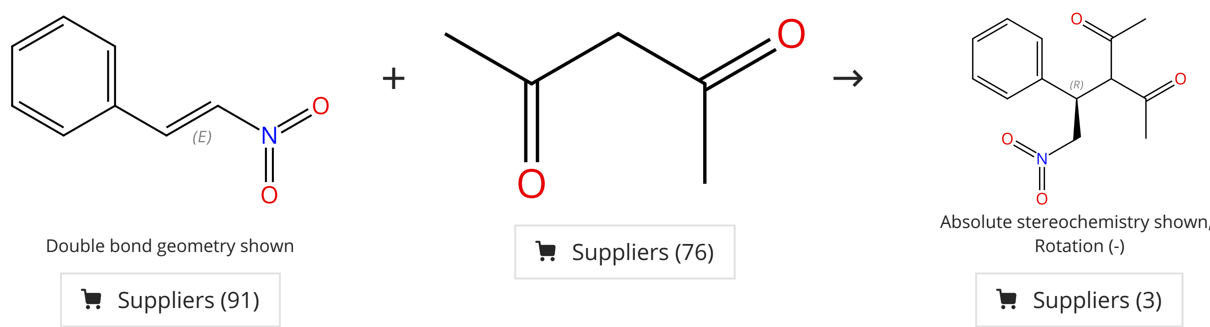
Experimental Protocols

Chiral-at-Metal Ruthenium Complexes with Guanidinobenzimidazole and Pentaphenylcyclopentadienyl Ligands: Synthesis, Resolution, and Preliminary Screening as Enantioselective Second Coordination Sphere Hydrogen Bond Donor Catalysts

By: Mukherjee, Tathagata; et al

Organometallics (2020), 39(8), 1163-1175.

Scheme 47 (1 Reaction) Steps: **1** Yield: **99%**



31-614-CAS-37446688 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Sorbitol, Betaine, Water, 3-[[[3,5-Bis(trifluoromethyl)phenyl]methyl]amino]-4-[(8α,9*S*)-cinchonan-9-ylamino]-3-cyclobutene-1,2-dione; 24 h, 40 °C

Experimental Protocols

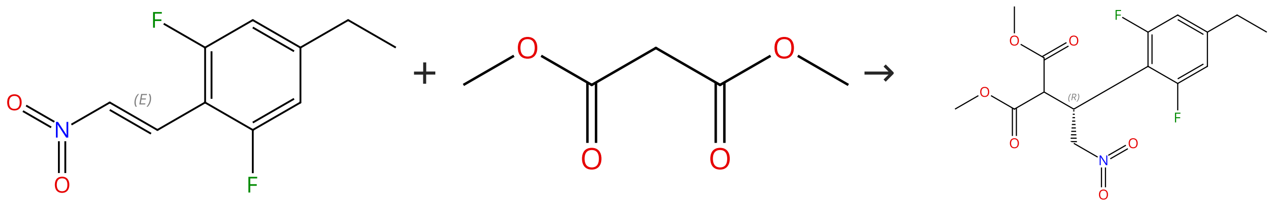
Sustainable Organocatalyzed Enantioselective Catalytic Michael Additions in Betaine-Derived Deep Eutectic Solvents

By: Fonseca, Daniela P.; et al

SynOpen (2023), 7(3), 374-380.

Scheme 48 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (90)

Absolute stereochemistry shown

Supplier (1)

31-254-CAS-22515541 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*OC*-6-12)-Bis[(1*S*,2*S*)-*N*[1],*N*[2]-bis(phenylmethyl)-1,2-cyclohexanediamine-κ*N*[1],κ*N*[2]]dibromonickel
Solvents: Toluene; 51 h, rt

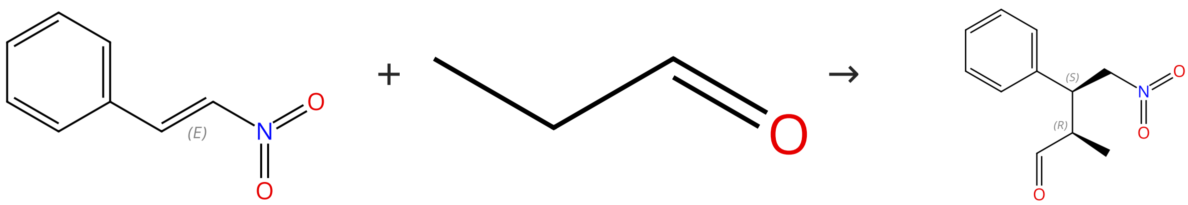
Discovery of BMS-986235/LAR-1219: A Potent Formyl Peptide Receptor 2 (FPR2) Selective Agonist for the Prevention of Heart Failure

By: Asahina, Yoshikazu; et al

Journal of Medicinal Chemistry (2020), 63(17), 9003-9019.

Scheme 49 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (91)

Suppliers (75)

Absolute stereochemistry shown, Rotation (+)

Suppliers (2)

31-614-CAS-30638003 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 11-[4-[[[(3*R*,5*S*)-5-[[[(1,1-Dimethylethyl)dimethylsilyl]oxy]diphenylmethyl]-3-pyrrolidinyl]oxy]methyl]-1*H*-1,2,3-triazol-1-yl]-1,1-undecanediol (calcium carbonate-supported)
Solvents: Dichloromethane; 24 h, rt

1.2 rt

Experimental Protocols

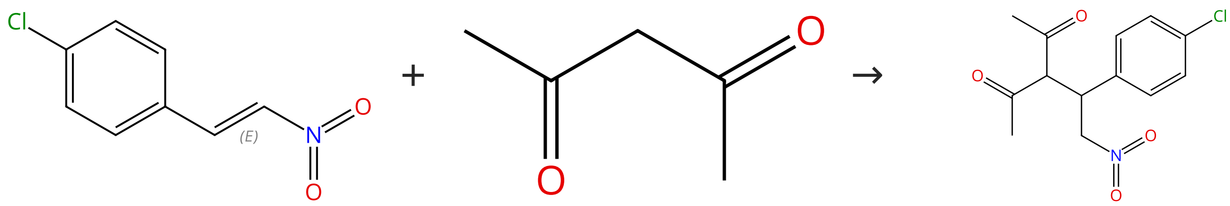
Calcium carbonate as heterogeneous support for recyclable organocatalysts

By: Lizandara-Pueyo, Carlos; et al

Journal of Catalysis (2021), 393, 107-115.

Scheme 50 (1 Reaction)

Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (57)

Suppliers (76)

Supplier (1)

31-254-CAS-21265688 Steps: **1** Yield: **99%**

1.1 **Catalysts:** 2377799-78-3
Solvents: Ethanol; 16 h, rt

Experimental Protocols

Molecular Engineering of β-Substituted Oxoporphyrinogens for Hydrogen-Bond Donor Catalysis

By: Chahal, Mandeep K.; et al

European Journal of Organic Chemistry (2020), 2020(1), 82-90.